

Supporting information: Size-selective optical printing of silicon nanoparticles through their dipolar magnetic resonance

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Optical forces on a Silicon sphere located at the focal point of a focused dipolar beam

1. Plane wave excitation

For completeness, let us first discuss the forces exerted by a circularly polarized plane wave with helicity $\sigma = \pm 1$ (we associate left polarized light with a positive helicity $\sigma = +1$). In a homogeneous medium with real constant permittivity ϵ_h (and refractive index $n_h = \sqrt{\epsilon_h}$) the multipole expansion of the circularly polarized plane wave is given by (see for example [1])

$$\mathbf{E}_\sigma = E_0 \frac{\hat{\mathbf{x}} + \sigma i \hat{\mathbf{y}}}{\sqrt{2}} e^{ikz} = E_0 \sum_{l=1}^{\infty} i^l \sqrt{2\pi(2l+1)} \left[j_l(kr) \mathbf{X}_{l\sigma} + \sigma \frac{\nabla \times [j_l(kr) \mathbf{X}_{l\sigma}]}{k} \right] \quad (\text{S1})$$

(where $k = 2\pi n_h/\lambda$), while for a plane wave linearly polarized along $\hat{\mathbf{x}}$ or $\hat{\mathbf{y}}$ is simply given by

$$\mathbf{E}_x = \frac{E_- + E_+}{\sqrt{2}} = E_0 e^{ikz} \hat{\mathbf{x}}, \quad \text{or,} \quad \mathbf{E}_y = i \frac{E_- - E_+}{\sqrt{2}} = E_0 e^{ikz} \hat{\mathbf{y}}. \quad (\text{S2})$$

The optical force exerted by the plane wave on a homogenous dielectric sphere of radius R is proportional to the radiation pressure cross section and is given by [2]:

$$\mathbf{F}_z = \hat{\mathbf{z}} \frac{\epsilon_0 \epsilon_h}{2} \pi R^2 |E_0|^2 \left[\frac{6}{(kR)^2} \text{Re}\{a_1 + b_1 - a_1 b_1^*\} + \dots \text{"higher order multipoles"} \right] \quad (\text{S3})$$

where a_1 and b_1 are the electric and magnetic dipolar ($l=1$) Mie coefficients, respectively. The radiation pressure does not depend on the helicity of the incoming plane wave. Notice that the radiation pressure induced by the plane wave contains not only dipolar but also higher order multipolar contributions.

2. Dipolar focused beam

The field of any focused beam around the focus can be expressed as a sum of multipolar vector wave functions. For simplicity, we consider the limit of a “dipolar” electric and magnetic focused beam with well-defined helicity σ whose field is identical to the first term of the expansion of a circularly polarized plane wave around the origin (*e.g.*, the term $l=1$ in Eq. (S2))

$$\mathbf{E}_\sigma = E_0 [i\sqrt{6\pi}] \left[j_1(kr) \mathbf{X}_{1\sigma} + \sigma \frac{\nabla \times [j_1(kr) \mathbf{X}_{1\sigma}]}{k} \right] \quad (\text{S4})$$

(This corresponds to Equation 1 in the main text). Analogously, we can construct a linearly polarized dipolar focused beam as

$$\mathbf{E}_x = \frac{\mathbf{E}_- + \mathbf{E}_+}{\sqrt{2}} = E_0 [i\sqrt{3\pi}] \left[j_1(kr) (\mathbf{X}_{11} + \mathbf{X}_{1-1}) + \frac{\nabla \times [j_1(kr) (\mathbf{X}_{11} - \mathbf{X}_{1-1})]}{k} \right] \quad (\text{S5})$$

The intensity profiles on the focal plane of both circularly and linearly polarized beams are shown in Fig. S1.

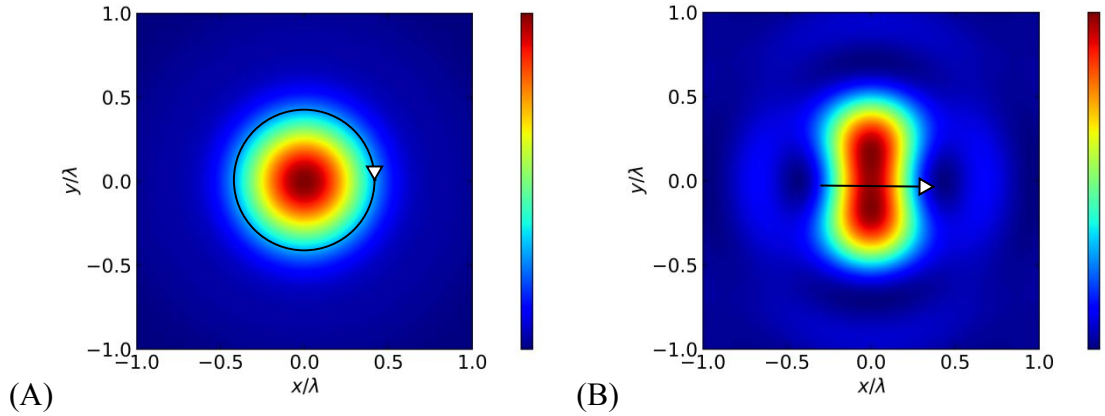


Figure S1: Field intensity map, $|\mathbf{E}|^2$, on the focal plane ($z=0$) for a circularly polarized (A) and for a linearly polarized (B) dipolar beams. Arrows represent the actual field polarization at $\mathbf{r}=0$. Although the axial symmetry of the beam is lost for linear polarization, the actual radiation pressure force on a spherical particle centered at $\mathbf{r}=0$ is the same.

When the sphere center is located at the origin, i.e., in the focus, the dipolar beam can only excite electric, $\mathbf{p}$, and magnetic, $\mathbf{m}$, dipoles on the sphere, independently of the particle size. In terms of the Mie coefficients,

$$\mathbf{p} = \varepsilon_0 \varepsilon_h i \left(\frac{6\pi}{k^3} \right) a_1 \mathbf{E}(\mathbf{r} = 0) ; \quad \mathbf{m} = i \left(\frac{6\pi}{k^3} \right) b_1 \mathbf{H}(\mathbf{r} = 0)$$

3. Optical forces and radiation pressure efficiency

The optical force on a (electric and magnetic) dipolar sphere, located at $\mathbf{r} = 0$, is given by [3]:

$$\begin{aligned} \mathbf{F}_z &= \hat{z} \frac{1}{2} \text{Re} \left\{ \mathbf{p}^* \cdot \frac{\partial}{\partial z} \mathbf{E} + \mu_0 \mathbf{m}^* \cdot \frac{\partial}{\partial z} \mathbf{H} - \frac{Z k^4}{6\pi} (\mathbf{p} \times \mathbf{m}^*) \cdot \hat{z} \right\}_{\mathbf{r}=0} = \\ &= \hat{z} \frac{\varepsilon_0 \varepsilon_h}{2} \pi R^2 \frac{6}{(kR)^2} \text{Re} \left\{ a_1 \frac{i \mathbf{E}}{k} \cdot \frac{\partial \mathbf{E}^*}{\partial z} + b_1 \frac{i Z^2 \mathbf{H}}{k} \cdot \frac{\partial \mathbf{H}^*}{\partial z} - a_1 b_1^* Z (\mathbf{E} \times \mathbf{H}^*) \cdot \hat{z} \right\}_{\mathbf{r}=0} \end{aligned} \quad (\text{S6})$$

where $Z = n_h / (\varepsilon_0 \varepsilon_h c) = \sqrt{\mu_0 / (\varepsilon_0 \varepsilon_h)}$ and $Z \mathbf{H} = -i \nabla \times \mathbf{E} / k$.

From Eq. (S4) we obtain

$$\left\{ \frac{i \mathbf{E}_\sigma}{k} \cdot \frac{\partial \mathbf{E}_\sigma^*}{\partial z} \right\}_{\mathbf{r}=0} = \left\{ \frac{i Z^2 \mathbf{H}_\sigma}{k} \cdot \frac{\partial \mathbf{H}_\sigma^*}{\partial z} \right\}_{\mathbf{r}=0} = \frac{1}{2} |E_0|^2 \quad (\text{S7})$$

$$Z (\mathbf{E}_\sigma \times \mathbf{H}_\sigma^*) \cdot \hat{z} = |E_0|^2 \quad (\text{S8})$$

(notice that the result of Eq. (S7) is half of the one obtained for a plane wave) and the force is then given by

$$\mathbf{F}_z = \hat{z} \frac{\varepsilon_0 \varepsilon_h}{2} \pi R^2 |E_0|^2 \left[\frac{6}{(kR)^2} \text{Re} \left\{ \frac{a_1 + b_1}{2} - a_1 b_1^* \right\} \right] \quad (\text{S9})$$

which leads to the radiation pressure efficiency, Eq. (2) in the main text,:

$$Q_{pr} = \frac{2}{(kR)^2} \left[3 \text{Re} \left(\frac{a_1 + b_1}{2} - a_1 b_1^* \right) \right] \quad (\text{S10})$$

For linearly polarized beams, e.g. Eq. (S5), we obtain exactly the same results as in Eqs. (S7) and (S8). When the particle is located at the focal point, $\mathbf{r} = 0$, the radiation pressure does not depend on the polarization of the dipolar beam.

Figure 2C in the main text shows Q_{pr} calculated from Eq. (S10) assuming a constant refractive index $n_h = \sqrt{\epsilon_h} \sim 1.33$ for water and Silicon refractive index (n_{Si}) data obtained from Ref. [4]. It is interesting to note that Q_{pr} is a function of the size parameter $kR = 2\pi n_h R / \lambda$ and the relative refractive index n_{Si}/n_h .

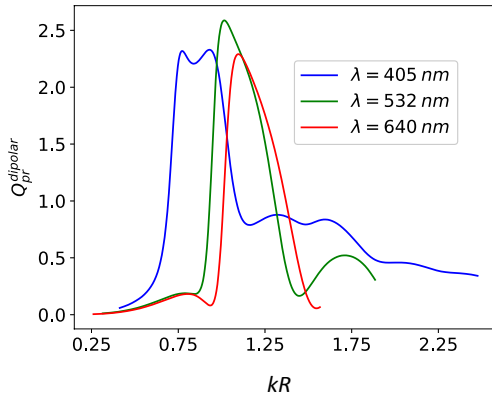
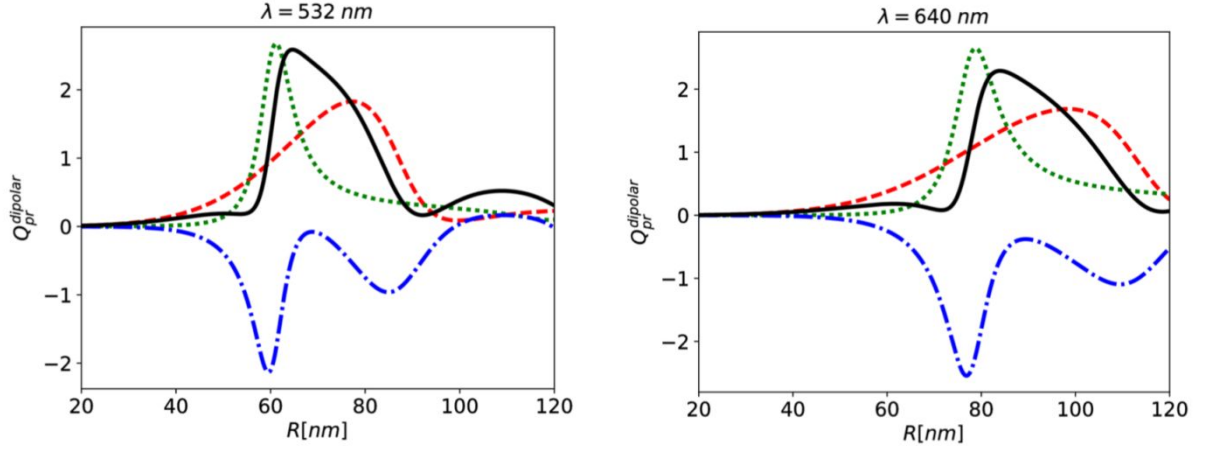


Figure S2: Radiation pressure efficiency Q_{pr} vs. size parameter kR at three selected wavelengths used in the experiments: 405 nm, 532 nm, and 640 nm computed using experimental refractive index from Ref. [4]

Figure S2 shows the same Q_{pr} as in Fig. 2C but now plotted versus kR . As it can be seen, Q_{pr} presents a maximum value of ~ 2.5 at $kR_0 \sim 1$, which can be attributed to the magnetic dipolar mode. However, the observed behavior results from a non-trivial balance of three different contributions as shown in Fig. S3.



--- $3\text{Re}\{a_1\}/(kR)^2$
 $3\text{Re}\{b_1\}/(kR)^2$
 -.-. $-6\text{Re}\{a_1 b_1^*\}/(kR)^2$
 — Q_{pr}

Figure S3: Radiation pressure efficiency Q_{pr} vs. radius R (continuous black line) is given by the sum of three different contributions (Eq. S10). The maximum of Q_{pr} is similar to the resonant magnetic contribution (dotted green line).

Results obtained at different wavelengths are different due to the different particle's refractive index

$$\frac{n_{Si}(\lambda = 405)}{n_h} = 4.079 + i0.249; \frac{n_{Si}(\lambda = 532)}{n_h} = 3.120 + i0.039; \frac{n_{Si}(\lambda = 640)}{n_h} = 2.909 + i0.013.$$

The scaling properties of Q_{pr} are illustrated in Figure S4A where we assume a constant refractive index $n_{Si}/n_h = 3.120 + i0.039$ for all the wavelengths. The actual maximum of Q_{pr} depends on the interplay between real and imaginary (absorption) parts of the Si refractive index. The influence of the absorption is shown in Figure S4B where we assume a real refractive index $n_{Si}/n_h = 3.120$.

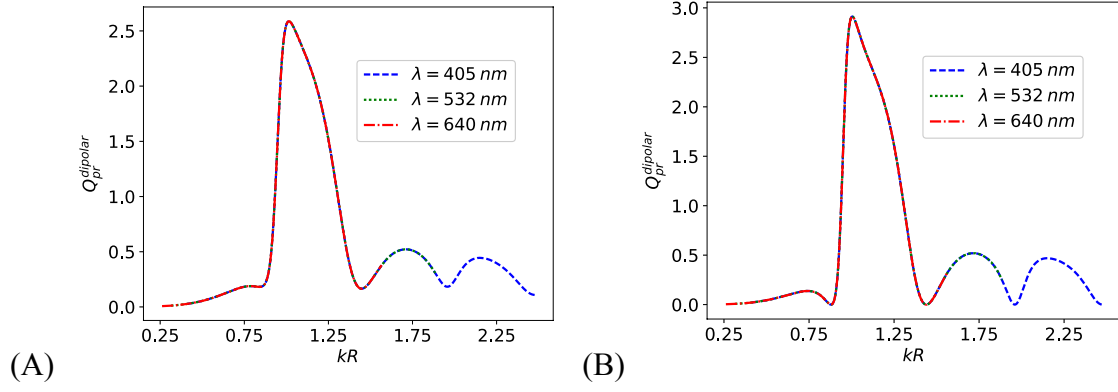


Figure S4: Radiation pressure efficiency Q_{pr} vs. size parameter kR at three selected wavelengths used in the experiments: 405 nm, 532 nm, and 640 nm assuming **(A)** constant refractive index $n_{Si}/n_h = 3.120 + i0.039$ for all the wavelengths and **(B)** “real” constant refractive index $n_{Si}/n_h = 3.120$ for all the wavelengths.

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